

🍔 Exploratory Data Analysis (EDA)

📌 Project Overview

This Exploratory Data Analysis (EDA) was performed to understand McDonald's transactional sales data before building the interactive Excel dashboard.

The objective was to clean the data, identify sales patterns, analyze customer purchasing behaviour, and prepare a well-structured data model suitable for business reporting.

📁 Dataset Description

The project consists of two datasets.

1. Menu Items

Contains product information.

Column	Description
-----	-----
menu_item_id	Unique menu identifier
item_name	Product name
category	Product category
price	Selling price

2. Order Details

Contains customer transaction records.

Column	Description
order_details_id	Transaction identifier
order_id	Order identifier
order_date	Order date
order_time	Order time
item_id	Purchased menu item

Additional columns were generated during data preparation.

- Price
- Day Name
- Workday Type
- Time Period

🛠 Data Cleaning

Power Query was used for data preparation.

Cleaning steps included:

- Removed blank rows
- Removed unnecessary columns
- Corrected data types
- Formatted price as Currency
- Checked duplicate records
- Handled missing values
- Merged menu and order datasets using ****menu_item_id****
- Verified relationship integrity

Feature Engineering

Several calculated columns were created to support analysis.

Day Name

Extracted weekday names from ****order_date****.

****Examples****

- Monday
- Tuesday
- Wednesday

****Used for:****

- Daily sales analysis

- Weekday vs Weekend comparison

Workday Type

Orders were classified into:

- Weekday
- Weekend

Purpose

To compare customer purchasing behaviour between working days and weekends.

Time Period

Orders were grouped into four business-friendly intervals.

| Time | Period |

|-----|-----|

| 6 AM – 12 PM | Morning |

| 12 PM – 5 PM | Afternoon |

| 5 PM – 9 PM | Evening |

| 9 PM – 6 AM | Night |

This enables customer traffic analysis throughout the day.

📁 Data Modeling

A ****Star Schema**** was implemented.

Dimension Table

****Menu_items****

Fact Table

****order_details****

Relationship

``text

Menu_items[menu_item_id]

|

| One-to-Many



order_details[item_id]

``

Cardinality

****One-to-Many****

Benefits

- Faster PivotTables
- Reduced redundancy
- Better dashboard performance
- Easier filtering using slicers

Univariate Analysis

The following variables were explored individually.

Sales

Sales values were analyzed to understand revenue distribution across products.

Categories

Menu items belong to categories such as:

- Burger
- Chicken
- Breakfast

- Pasta
- Fries
- Sandwich
- Wrap
- Drinks
- Sides
- Salad
- Shake

Orders

Order volume was inspected to identify customer demand.

Bivariate Analysis

Relationships between variables were explored.

Category vs Sales

Compared total sales generated by each category.

****Observation****

- Burger category generated the highest revenue.

Time Period vs Orders

Compared customer order activity across:

- Morning
- Afternoon
- Evening
- Night

****Observation****

- Afternoon recorded the highest customer activity.

Weekday vs Weekend

Sales were compared between weekdays and weekends.

****Observation****

- Weekday sales consistently exceeded weekend sales.

Product vs Revenue

Products were ranked according to revenue contribution.

****Observation****

- Top-performing products contributed significantly to total sales.

📅 Trend Analysis

Daily Trend

Daily order volume remained relatively consistent with normal fluctuations.

Monthly Trend

Monthly revenue remained stable with periodic increases during several days.

Category Trend

Burger and Chicken categories consistently outperformed other menu segments.

📊 Dashboard KPIs

The dashboard summarizes overall business performance.

KPI Value
----- -----:
Total Sales **\$61,003.78**
Total Orders **5,370**
Average Order Value (AOV) **\$11.36**
Total Quantity Sold **12,234**

💡 Key Business Insights

- 🍔 Burgers generated the highest sales among all menu categories.
- ☀️ Afternoon experienced the highest customer traffic.
- 📅 Weekday sales outperformed weekend sales.
- 💰 Premium and budget-friendly products were identified using menu prices.
- 📊 Daily sales remained stable with moderate fluctuations.
- 📈 Monthly revenue analysis revealed consistent business performance.
- 🔑 Category-level analysis highlighted Burger and Chicken as major revenue contributors.

📌 Business Recommendations

- Promote premium products during afternoon peak hours.
- Increase promotional offers on weekends to improve weekend sales.
- Bundle lower-performing menu items with best-selling products.
- Monitor daily sales to identify seasonal demand patterns.
- Introduce targeted marketing campaigns for slower categories.

🚀 Conclusion

The exploratory analysis successfully transformed raw transactional data into structured business information. The cleaned dataset, engineered features, and optimized Star Schema enabled the creation of an interactive Excel dashboard that provides valuable insights into sales performance, customer behaviour, and product performance.